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Beyond Porosity: The Fractality of Grain Size Distributions and Rock Property Prediction

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Abstract

The objective of this work is to help enable image-based predictions of mechanical properties while contributing to the general understanding of the microstructural drivers for stiffness and strength in natural aggregates such as sedimentary rocks. Initial results on an investigation into the relationship between internal structure and elastic stiffness are presented. Stiffness values are computed using commercial simulation software on 3D X-ray CT images of Boise sandstone, Fontainebleau sandstone, as well as on synthetic media comprising a spherical grain pack and a homogeneous material containing spherical inclusions. In order to isolate the structural component, a monomineralic composition is assumed with identical solid elastic properties throughout. Image volumes exhibiting the same porosity, but different computed stiffness values are extracted for further study. The volumes are analyzed with a morphological image processing tool that measures local thickness, and which is used as a proxy for material size distribution within the rock skeleton. The resulting distributions are compared within the framework of grading curves as is routinely done in other engineering domains. The results indicate possible fractal characteristic for the size distributions, with the fractal dimension visually ranking with stiffness for a fixed porosity. Further work is warranted to confirm these results via improved image processing.

Introduction

Whereas porosity is known as a fundamental driver for aggregates stiffness and strength, the influence of other factors such as mineralogy, grain or pore sizes and shapes, etc. is not as easily quantifiable. Grain size distribution (GSD) has been the focus of a number of studies from concrete manufacturing to natural fault gouge formation, and compaction localization in high porosity oil and gas reservoirs. GSD, or sorting, is known to influence physical properties, though it tends correlate strongly with porosity and hence does not contribute as a purely independent factor (Florez-Nino, 2005). Poor sorting (i.e. a wide grain size distribution) offers opportunities for ever smaller intergranular spaces to be filled with ever smaller grains. The principle notably applies during comminution, where grains are mechanically crushed under compressive stress in order to achieve permanent volume reduction. Throughout this process, and regardless of initial sorting, GSDs tend to converge towards power laws with well-defined fractal dimensions (Turcotte, 1986; Steacy and Sammis, 1991; Ben-Nun et al., 2010; Yang et al., 2022).

Beyond the statistical aspect, however, by constraining the number and dimensions of achievable intergranular contacts, GSDs also influence the overall architecture of an aggregate, and hence the transmission of mechanical forces. Assuming two porous aggregates of identical porosity and composition, but exhibiting distinct elastic stiffnesses, what precisely in their respective structures is responsible for the observed difference, and more importantly, how to measure it so it can be used predictively? Such a parameter, which should be tied to the grains' dimensions and spatial arrangement, could potentially bridge granular and inclusion-based approaches, which are currently considered separately with their respective sets of models.

In this paper, we initiate a systematic image-based analysis aimed at probing the effect of internal structure on calculated elastic stiffness. The main image data set consists of an X-ray CT volume of a Boise sandstone imaged at a resolution of 3.425 microns per voxel. Additional images are also used for the study, which are an X-ray CT volume of Fontainebleau sandstone obtained at 5.7 microns per voxel (Louis et al., 2007; Lindquist et al., 2000), and three synthetic image sets: a series of granular packings of monodisperse spheres, and two series of solid cubes, each with monodisperse spherical inclusions. From these volumes, static elastic tensors are first computed with a commercial simulation software (Andrä et al., 2013). The results are then compared against digital porosity to establish a baseline and assess the presence of other potential microstructural controls. Small simulation volumes of identical porosity but with distinct computed stiffness values are extracted for further analysis. Using a maximal inscribed sphere (or local thickness) algorithm (Hilpert and Miller, 2001; Silin and Patzek, 2006; Dougherty and Kunzelmann, 2007) as a proxy for probing the dimensions of the solid fraction, we visually compare grain size statistics with stiffness values. In the discussion, the results are weighed against the limitations of the approach used, and elements for future work are outlined which should enable a better understanding and predictability of mechanical properties in general from 3D images.

Previous Work

The present study expands on the work conducted in the general area of local strength prediction in granular materials. This is especially important when trying to predict compactive strain localization, but applies to static and dynamic elastic properties as well. In the case of localized compaction and its propagation, studies have pointed to porosity as a main control, then to local 'heterogeneity' as a secondary one, where heterogeneity tends to hinder propagation, whereas homogeneity promotes it (Baud et al., 2004; Louis et al., 2009a-b; Wong and Baud, 2012; Cheung et al., 2012; Baud et al., 2015; Carbillat et al., 2022). The specifics of what is behind this concept of heterogeneity are ill-defined and may comprise mineralogy, texture, pore or grain size distribution, etc. In a recent study, Louis et al. (2024) argued that if grain comminution during compaction leads to fractal GSDs as this provides the most compact arrangement, perhaps the proneness to yielding of an aggregate could be related to how distant its GSD is from that ideal configuration. In a continuation of these investigations, we attempt here to connect elastic stiffness with GSD, independently of the porosity.

Materials and Methods

In Fig. 1, all the types of material on which numerical simulations were performed are presented. Fig. 1a shows the main image volume which was obtained in Boise sandstone at a resolution of 3.542 microns per voxel. The volume was resized to half that resolution, bringing it to 6.85 microns per voxel in order to optimize image handling and computations. A script was written in ImageJ (Schneider et al., 2012) to extract all contained 150 voxel-wide cubes, with an offset of 100 voxels, to probe for and pick domains with a variety of porosities and structures. In addition, four 150 voxel-wide volumes were extracted from the Fontainebleau data set studied by Louis et al. (2007) (Fig. 1b); one monodisperse grain pack volume was generated using a commercial software module then sequentially eroded to produce different porosity

domains; and finally another script was written to generate two monodisperse spherical inclusion domains with a ratio of 2 between diameters for the two cases, then sequentially eroded to produce different porosity domains as was done for the grain pack. All simulated volumes share the same dimensions of 150^3 vox^3 .

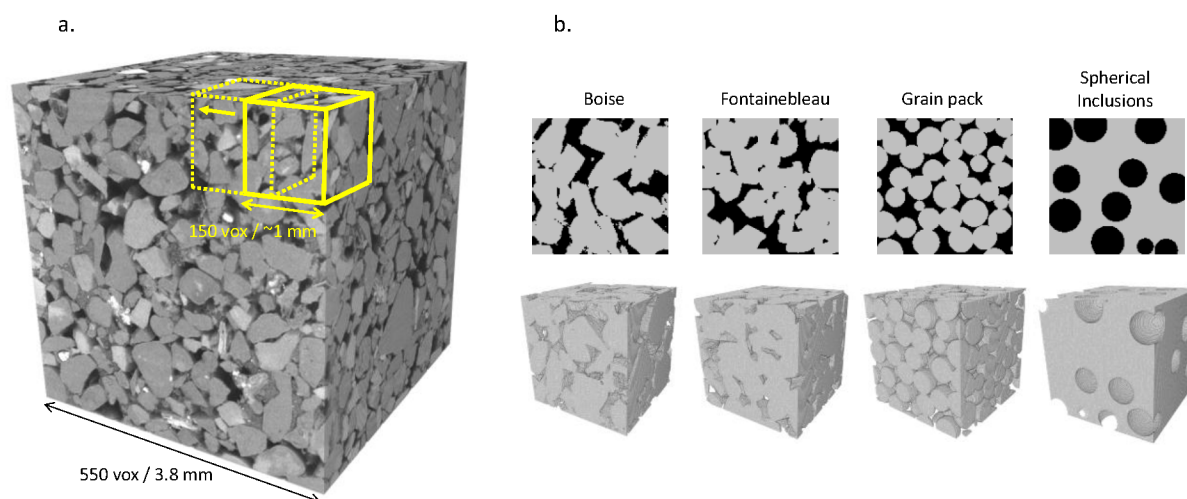


Figure 1—Image volumes used for the elastic computations. (a) Initial image volume of Boise sandstone and sub-volume extraction strategy; (b) Overview of domain size and structure for all types of materials studied.

When not already in binary format, image volumes were segmented using the default ImageJ Otsu thresholding method which minimizes internal variance within phases. Then the segmented volumes were cleaned to remove floaters (i.e. small grains not connected to the main skeleton) as well as trapped porosity.

For this work, a total of 56 volumes were simulated (37 Boise, 4 Fontainebleau, 5 grain pack, 6 large inclusions, 4 small inclusions) using available digital rock commercial software to obtain an orthorhombic elastic tensor, which was then averaged to the closest set of isotropic elastic bulk (K) and shear (G) moduli. Material properties used for the solid fraction were those of Quartz ($K=36\text{GPa}$, $G=45\text{GPa}$), and the pore space was assumed to be empty ($K=0\text{GPa}$, $G=0\text{GPa}$).

Results

Simulation Results

In Fig. 2, all values for K and G are plotted against porosity. Two main trends arise, with granular materials on one and inclusions on the other. Scatter in Boise sandstone data show significant variability among domains coming from a unique 3.8 mm-wide cube. In this case, as the simulations are sensitive to porosity and structure only, Boise and Fontainebleau sandstones share the same trend, even though large differences would be expected in actual laboratory measurements due to the presence in the Boise sandstone of soft minerals and compliant intergranular contacts, while the Fontainebleau is almost made up of 100% quartz with well cemented grain contacts. From a structural point of view, the inclusion samples are much stiffer than the granular ones as the latter exhibit constriction at the intergranular contacts, lowering the overall contacting surface and concentrating stresses, which in turn cause larger strains and lower computed stiffnesses. In Fig. 2c, a zoomed in section of Fig. 2a shows a group of 7 samples (red dotted rectangle) with very similar porosity, but for which the bulk modulus varies by as much as 9 GPa (or ~50%) for all volumes and 3 GPa (or ~20%) within the Boise ones. The porosity and moduli values for these seven data points are provided in Tab. 1. This observation clearly illustrates the question that this work aims to address, i.e. what metric can be found in the image of the solid fraction that could explain and help quantify the influence of the skeleton's structure/morphology on elastic and mechanical properties beyond porosity? Answering this question appears essential as the effect of structure is very substantial, should logically

precede the discussion on mixing laws for multicomponent aggregates, and could also provide a new avenue to modeling changes in properties upon burial and compaction, where morphology is reworked at virtually constant mineralogy.

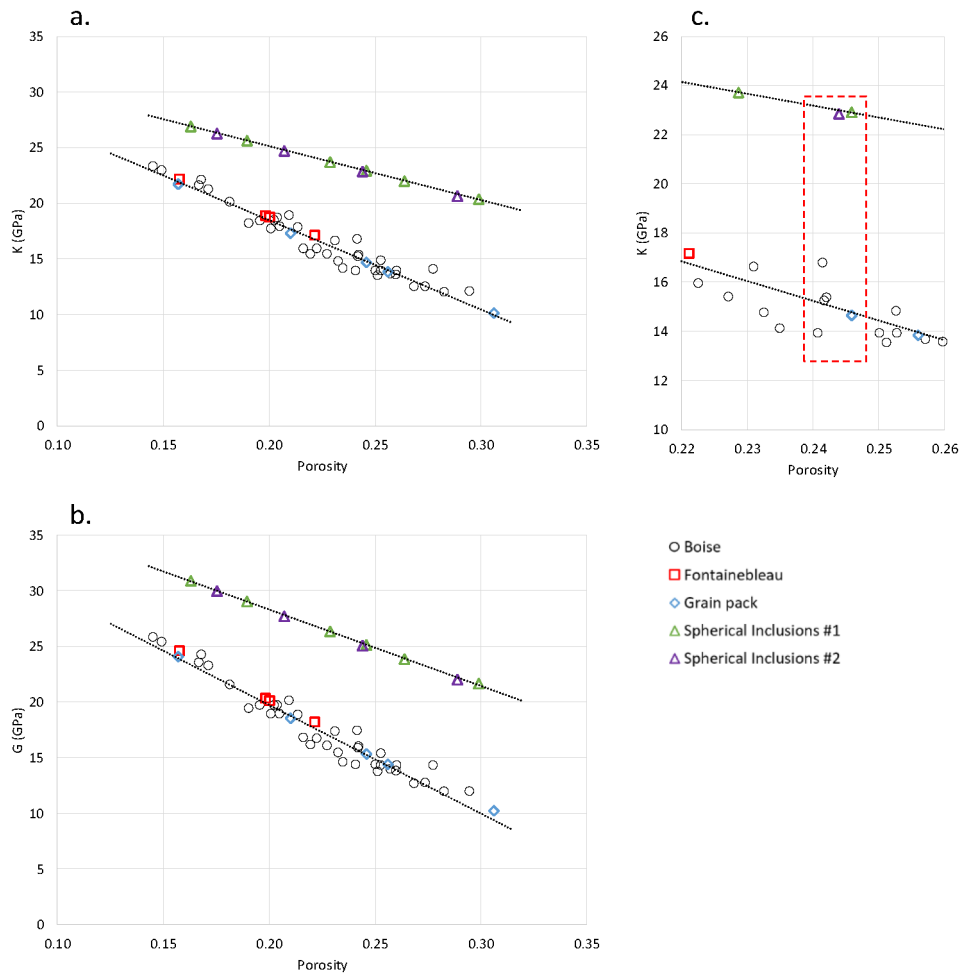


Figure 2—Elastic moduli vs. porosity for all simulations. (a) Bulk modulus as a function of porosity. (b) Shear modulus as a function of porosity. (c) Zoomed-in portion of (a) showing results variability at ca. 24% porosity.

Table 1—Porosity and moduli values for the seven samples in the red box of Fig. 1c.

Sample ID	Porosity	K (GPa)	G (GPa)
#35	0.242	15.36	15.97
#54	0.242	16.79	17.39
#70	0.242	15.24	15.79
#107	0.241	13.9	14.3
GP4	0.246	14.64	15.25
INC1	0.246	22.9	25.1
INC2	0.244	22.85	25.04

Image Analysis and Interpretation

To initiate the search for a microstructural indication of structure within the solid skeleton, we adopt the use of a maximal inscribed sphere (MIS) algorithm as a proxy for grain size distribution, as was previously proposed in Louis et al. (2016, 2024), although this only strictly applies to sphere packs. Also called local thickness, the algorithm is one way of probing the internal structure of the skeleton.

In Fig. 2, four different MIS realizations are presented. In Figs. 2a and 2b, granular assemblies produce predictable patterns, except for a few artefacts in the grain pack coming from the bounding surfaces. In

Figs. 2c and 2d, though the skeleton is not composed of grains per se, the tool is still indicative of local dimensions distribution.

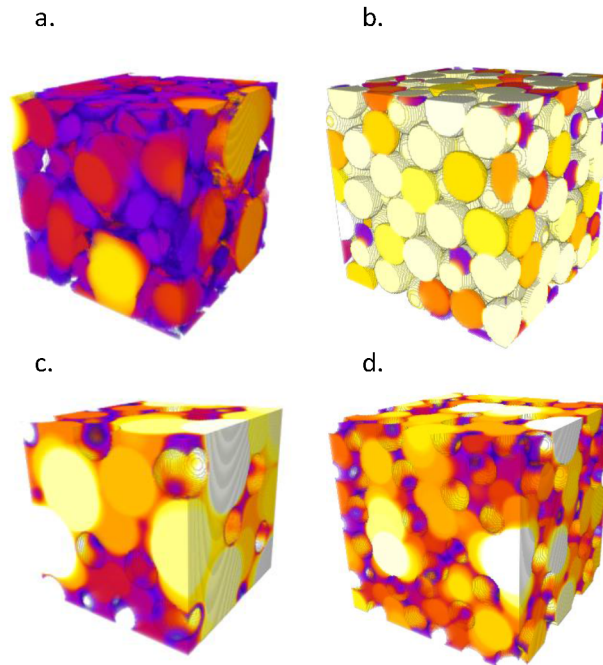


Figure 3—Different MIS realizations for the types of materials tested. (a) Boise sandstone; (b) Spherical grain pack; (c) Spherical inclusions #1; (d) Spherical inclusions #2.

Studies involving granular aggregates in hydraulic, civil or transport engineering often analyze GSDs in terms of volumes rather than numbers, using terms such as percent passing, fractional volume, or fractional mass, leading to the concept of grading curves (Yang et al., 2022; Kalore and Babu, 2023). A very particular case of these distributions is the one of a fractal distribution, such as emerges through crushing, fragmentation or blasting. While not directly relevant in the present study since none of the materials simulated has undergone crushing, power laws in general can be used to canvas grading curves, with one particular end-member being the Fuller curve (Fuller and Thompson, 1907). The Fuller curve is an empirically determined grading curve of exponent 0.5 (and fractal dimension D of 2.5), which correspond to the case of an ideal packing with nearly zero porosity (Borkovec et al., 1994). Grading curves are typically obtained by plotting a cumulative fractional mass against a normalized diameter ($d/d_{\max}$). In the case of fractal distributions, the grading curve can be expressed as:

$$F(d) = \left(\frac{d}{d_{\max}} \right)^{3-D} \quad (1)$$

where D represents the fractal dimension.

The MIS tool used here inherently produces volumetric data since a single sphere will be represented in a histogram by all the voxels inscribed within that sphere. We show in Fig. 4 the cumulative MIS distributions for the seven sub-volumes. In Fig. 4a, original diameter values are used, which is not a good gauge for comparing samples as we are interested in relative contrasts independent of scale. In Fig. 4b., the normalized diameter values allow re-ranking of the grading curves. In Fig. 4c, the same curves are plotted in the log-log space, which better represents relationships across scales. The curves exhibit a sizeable linear (i.e. fractal) portion, with the ranking in slopes visually following the ranking between stiffnesses (red box). Distributions associated with the two spherical inclusion samples exhibit the steepest slopes. Within the Boise samples, the ranking corresponds to the simulation results with samples #107 (lowest values) and #54 (highest values) approximately bounding the range. The slope for the sand pack sample is comparable to

that of sample #107. While further investigations are needed, this initial result indicates that (1) fractality might be present in the grading curves of the materials studied and (2) higher fractal numbers at constant porosity might correlate with lower stiffness.

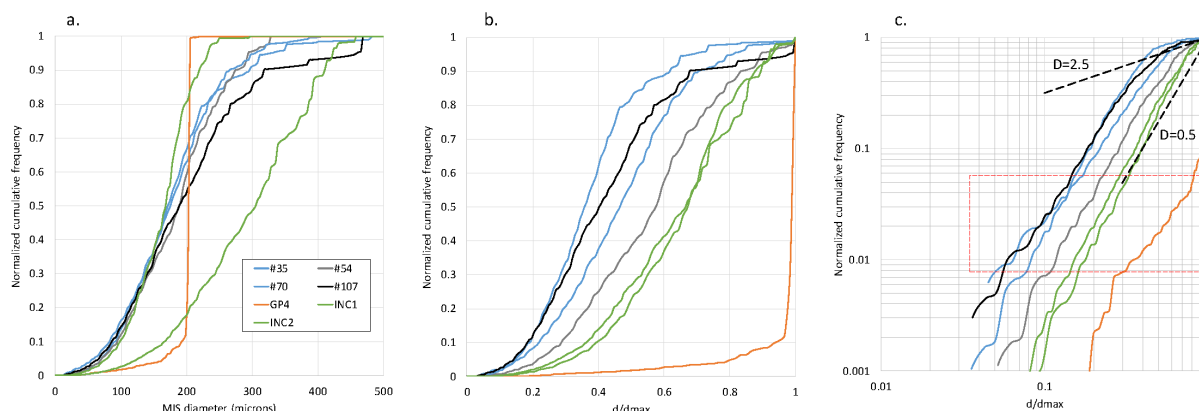


Figure 4—Grading curves obtained for the seven sub-volumes of interest in Tab. 1 and Fig. 2c. (a) Fractional volume vs. diameter; (b) Fractional volume vs. normalized diameter; (c) log/log version of (b).

Discussion and conclusion

The results obtained in this initial study are encouraging as they produce a potential differentiator among samples of similar porosity to explain different computed stiffness values. More work is however needed before a model can be put forth. In the case of the grain pack, the MIS analysis produced biased results which are likely responsible for the large imbalance in the grading curve. In Fig. 5, we show how an intergranular contact area, which should have exhibited a locally low thickness value due to a constriction, failed to do so as that area is seen as belonging to the adjacent spheres. In such cases, a neck at the interface would be needed for the constriction to be reflected. In the next steps of this investigation, a workaround will need to be applied in order to better reflect these critical geometric features. Also, while MIS statistics were used directly here, we did not investigate the full set of methods involving MIS and will perform digital intrusion porosimetry as proposed by Louis et al. (2016) to test for the presence of a controlling intergranular contact diameter, although the issue showed in Fig. 5 will still be at play and needs to be addressed first. Lastly, as the work progresses, the simulation data presented in Fig. 2 will be integrated with the microstructural interpretation in view of proposing a rock physics model parameterized with measurable geometric quantities.

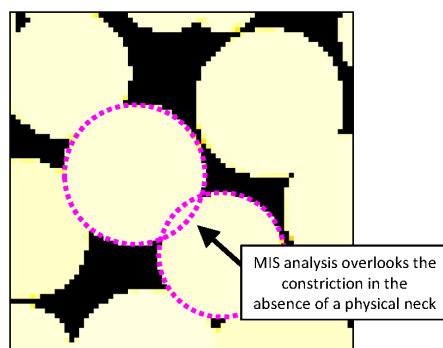


Figure 5—Demonstration of a shortcoming of the MIS analysis in the case of intersecting spherical domains. Here the true local thickness is not recognized as the contact area is seen as belonging to either of the spheres.

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